

- *maximal*: There are no intermediate rings strictly between A and its fraction field.

Various combinations of these properties can be used to uniquely characterize discrete valuation rings (and hence give alternative definitions).

Theorem 1.16. *For an integral domain A , the following are equivalent:*

- A is a DVR.
- A is a noetherian valuation ring that is not a field.
- A is a local PID that is not a field.
- A is an integrally closed noetherian local ring of dimension one.
- A is a regular noetherian local ring of dimension one.
- A is a noetherian local ring whose maximal ideal is nonzero and principal.
- A is a maximal noetherian ring of dimension one.

Proof. See [1, §23] or [2, §9]. □

1.6 Integral extensions

Integrality plays a key role in number theory, so it is worth discussing it in more detail.

Definition 1.17. Given a ring extension $A \subseteq B$, an element $b \in B$ is *integral over A* if it is a root of a monic polynomial in $A[x]$. The ring B is *integral over A* if all its elements are.

Proposition 1.18. *Let $\alpha, \beta \in B$ be integral over $A \subseteq B$. Then $\alpha + \beta$ and $\alpha\beta$ are integral over A .*

Proof. Let $f \in A[x]$ and $g \in A[y]$ be such that $f(\alpha) = g(\beta) = 0$, where

$$\begin{aligned} f(x) &= a_0 + a_1x + \cdots + a_{m-1}x^{m-1} + x^m, \\ g(y) &= b_0 + b_1y + \cdots + b_{n-1}y^{n-1} + y^n. \end{aligned}$$

It suffices to consider the case

$$A = \mathbb{Z}[a_0, \dots, a_{m-1}, b_0, \dots, b_{n-1}], \quad \text{and} \quad B = \frac{A[x, y]}{(f(x), g(y))},$$

with α and β equal to the images of x and y in B , respectively, since given any $A' \subseteq B'$ we have homomorphisms $A \rightarrow A'$ defined by $a_i \rightarrow a_i$ and $b_i \rightarrow b_i$ and $B \rightarrow B'$ defined by $x \mapsto \alpha$ and $y \mapsto \beta$, and if $x + y, xy \in B$ are integral over A then $\alpha + \beta, \alpha\beta \in B'$ must be integral over A' .

Let k be the algebraic closure of the fraction field of A , and let $\alpha_1, \dots, \alpha_m$ be the roots of f in k and let $\beta_1, \dots, \beta_n$ be the roots of g in k . The polynomial

$$h(z) = \prod_{i,j} (z - (\alpha_i + \beta_j))$$

has coefficients that may be expressed as polynomials in the symmetric functions of the α_i and β_j , equivalently, the coefficients a_i and b_j of f and g , respectively. Thus $h \in A[z]$, and $h(x+y) = 0$, so $x+y$ is integral over A . Applying the same argument to $h(z) = \prod_{i,j} (z - \alpha_i\beta_j)$ shows that xy is also integral over A . □